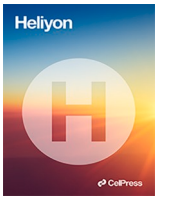
Heliyon 10 (2024) e25113



Contents lists available at ScienceDirect

Heliyon

journal homepage: www.cell.com/heliyon



Research article

Improvement of soil health and nutrient transformations under balanced fertilization with integrated nutrient management in a rice-wheat system in Indo-Gangetic Plains – A 34-year Research outcomes

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ARTICLE INFO

Keywords:
Farm yard manure
Green manuring
Wheat cut straw
Fixed and total K
Crystalline iron oxide
Carbonate
Amorphous iron oxide
Organic matter

ABSTRACT

The impact of integrated nutrient management seems crucial for the sustainability of crop production as revealed by studies on long-term experiments. It provided the opportunity to monitor long-term variations in crop yields and associated factors. The impacts of various nutrient management strategies on yields and soil attributes in a rice-wheat system have been researched under a long-term experiment that has been running since 1983 at Punjab Agricultural University, Ludhiana. Further, a positive correlation has been observed between crop yields and soil properties such as soil organic carbon (SOC), nitrogen (N), phosphorus (P), potassium (K) and zinc (Zn). The negative correlation with K could be attributed to soil becoming deficient in K and necessitating the application of potassium fertilizer. The treatments receiving organic manures (green manure, farmyard manure and wheat cut straw) showed a better population of soil microorganisms in comparison to the treatments receiving chemical fertilizers, thereby proving as precursors of sustaining soil health. The best soil characteristics (water-soluble aggregates, exchangeable and non-exchangeable K, fixed and total K) after rice and wheat harvesting were found where 50 % of the recommended NPK was supplemented with farmyard manure (FYM). The build-up of trace elements particularly for Fe and Zn was also noticed. In crystalline Fe oxide bound fraction (CFeOX), Fe increased between 717.1 and 984.8 mg kg^{-1}, while Zn increased between 2.64 and 3.08 mg kg^{-1}. Furthermore, amorphous iron oxide (AFeOX), CFeOX, carbonate (CARB), organic matter (OM) bound and exchangeable (EXCH) Fe and Zn were higher in

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https://doi.org/10.1016/j.heliyon.2024.e25113
Received 30 June 2023; Received in revised form 24 November 2023; Accepted 20 January 2024
Available online 3 February 2024
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treatments where organic manures were supplemented with 50 and 25 % N. Farmyard manure showed an incremental trend, followed by wheat cut straw and green manure (GM). The incremental trend in soil quality was noticed with FYM followed by wheat cut straw and GM.

1. Introduction

With the green revolution and adoption of improved agronomic practices, the rice-wheat cropping sequence (RWCS) has resulted in an impressive gain in production owing to high input use efficiency [1]. The system's input-intensive character makes it less economical as natural resources continue to deplete, such as the lowering of the water table and poor soil fertility [2]. On account of the exhaustive nature of this system, the productivity started declining on account of its multi-nutrient deficiency, the appearance of new bio-types, resistance in insect pests and weed species and its impact on soil physical properties caused by puddling [3]. Furthermore, the decadal trend study revealed a depletion of essential nutrients as well as a decrease in the quality and quantity of organic matter [4]. Another important finding explains that 22 and 11 % soils of Punjab are deficient in zinc (Zn) and iron (Fe), respectively [5] clearly advocating the necessity of customized fertilizers. Moreover, rice and wheat extract more potassium (K) from the soil than is present in the soil as available forms. The distribution of K in different chemical forms in the soil and their equilibrium defines the K status of soil and its potential to supply K to plants [6]. Even at the recommended rate of fertilisation, the potassium balance is negative because potassium additions rarely match potassium removals, resulting in a greater reliance on soil potassium. As a result, there is increased pressure on non-exchangeable K to meet crop requirements. The long-term intensive cropping in the absence of K inputs reduces K supply to crops, resulting in yield reduction [7].

The current situation justifies the necessity for the development of precision management techniques that can preserve soil health, keep the production system viable, and produce high-quality food to satisfy people's nutritional needs. The integrated nutrient management system (INM), enables us to incorporate a balanced form of the nutrients through combined use of organic and chemical sources of nutrition, which may be a reasonable idea for maintaining and managing long-term fertility and productivity of soil without having a detrimental effect on natural resources and environment. A key component of INM is maintaining soil fertility and crop productivity at the ideal to maximise the benefits of both inorganic and organic plant nutrient sources [8]. Organic matter management is the heart of sustainable agriculture. Crop residues and organic manure application affect the nutrient availability in soils [9]. It positively controls the uptake of nutrients and enhances soil quality, having a beneficial influence on crops [10,11]. For sustaining ideal crop yields and long-term soil productivity, organic manures and inorganic fertilisers should be used in conjunction [12]. The primary component of soil fertility is organic carbon. Soils that received both organic and inorganic fertilisers had higher levels of organic carbon [13,14]. Green manuring of *Sesbania rostrata* (a high N_2 fixer and stem nodulated legume) can increase the production of the following rice crop significantly and save between 50 and 60 per cent nitrogen. Its continuous use for three years can enhance soil's physico-chemical characteristics and have a considerable residual influence on the second crop i.e., wheat [15]. Ibrahim et al. [16] discovered that using Guar and *Sesbania* green manures in conjunction with NP considerably boosted the yield of both crops.

Thus, keeping in view the deteriorating soil health (nutritional disorder, degradation of soil physical properties, etc.), further coupled with the dynamics in physical, chemical and biological properties of soil, climate change, excessive use of water, and appearance of new biotypes, this study aims to develop and implement an INM approach to address these challenges and maintain the long term viability of the system.

2. Materials and methods

2.1. Description of the experimental site

A long-term field experiment was started in 1983–1984 and ran continuously until 2017–2018 on Typic Ustocrept soil at the Research Farm of Punjab Agricultural University, Ludhiana, at 30°56' N latitude and 75°52' E longitude with a mean elevation of 247 m above mean sea level. Agro-climatically, the experimental location is located in the Trans-Gangetic Plains Area, which has hot

Table 1
Physico-chemical characteristics of experimental soil.

Parameter	Values
Texture	Loamy sand
pH	8.15
Electrical conductivity	0.32 dS m^{-1}
Organic carbon	0.31 %
Available Nitrogen	143 kg ha^{-1}
Available Phosphorus	11.2 kg ha^{-1}
Available Potassium	101 kg ha^{-1}
DTPA-extractable Zn	1.96 mg kg^{-1}
DTPA-extractable Cu	0.80 mg kg^{-1}
DTPA-extractable Fe	9.80 mg kg^{-1}
DTPA-extractable Mn	9.14 mg kg^{-1}

summers, cool winters, and heavy monsoon rainfall. The experimental site receives 705 mm of rain annually on average, with 80 % of that falling during the monsoon season (June to September). The initial status of the experimental soil is given in Table 1.

2.2. Experimental design and treatments

The experiment was conducted with three replications in a randomized block design. The experiment included 14 different treatment combinations, which are listed in Table 2. These included applying nitrogen (N), phosphorus (P) and potassium (K) through fertilizers as separate components and combining these fertilizers with various organic sources to substitute 25 %–50 % of the N through farmyard manure (FYM), wheat cut straw (WCS) and green manure (GM). During the summer, different treatments were grown using farmyard manure, wheat crop residue, and Dhaincha (*Sesbania aculeate* L.), a crop planted as green manure. The recommended doses of N, P and K for rice were 105, 30, and 30 kg ha^{-1}, respectively. Before the last puddling, 33 % of the total N, full rates of P, K and 62.5 kg zinc sulphate ha^{-1} were applied. Three weeks after transplantation, one-third of the nitrogen was applied, and another one-third was applied six weeks later.

2.3. Soil sampling and laboratory analyses

After Rabi 2017–2018, plot-specific soil samples from surface soil (0–15 cm) from each replication were collected. The samples were cleaned, air-dried, and kept in polythene bags for analysis. The samples were examined for pH [17], soil organic carbon (SOC) content [18], available N [19], available P [20], and available K [21]. For microbial count, soil samples were taken from 0 to 15 cm soil layer at the harvest. Bacterial, actinomycetes and fungal populations of soil samples were estimated through the serial dilution plate count method by using Soil extract agar [22], Dextrose nitrate agar and Rose Bengal streptomycin agar medium [23]. Potassium was determined using a flame photometer for the available form of K [24], water-soluble K [25], fixed form of K [26], and non-exchangeable and total K [24]. Fig. 1 shows different fractions of Zn and Fe determined using the sequential fractionation method [27]. With an atomic absorption spectrophotometer, the Zn and Fe contents were measured.

2.4. Data analysis

To ascertain grain yield trends (slopes) over time, linear regression analysis was conducted. The P-value and t-statistics on the slopes were used to assess whether the observed changes were significantly different from 0 (P 0.05). To reduce the impact of unusually low or high yields in the first year of the experiment, the initial yield was assessed by the average of the first three years. The correlation analysis was carried out among different variables viz rice and wheat average yield, soil fertility status after rabi 2017-18 for OC, N, P and K and soil micronutrient status using SAS software.

3. Results and discussion

3.1. Effect of long-term INM on soil fertility status

The usage of organic sources of nutrition in the *kharif* season showed a discernible positive impact on soil SOC, soil N, available P and K contents and also the micro-nutrients.

3.1.1. Organic carbon

Soil OC content is the best indices of soil fertility influenced through the application of organic manures/crop residues/compost

Table 2
Different treatment combinations applied in the long-term experiment.

Treatments	Fertilizer use (% of recommended NPK)	
	Rice[a]	Wheat[a]
T1	Control	Control
T2	50	50
T3	50	100
T4	75	75
T5	100 (recommended dose)	100 (recommended dose)
T6	50 + 50 % N FYM	100
T7	75 + 25 % N FYM	75
T8	50 + 50 % N wheat cut straw	100
T9	75 + 25 % N wheat cut straw	75
T10	50 + 50 % N GM	100
T11	75 + 25 % N GM	75
T12	100 + 50 % N FYM	100
T13	$N_{180}P_{30}K_{30}$	$N_{150}P_{60}K_{30}$
T14	100	100+Cowpea in summer

[a] Rice: 105-30-30; N–P_2O_5–K_2O; kg ha^{-1} and wheat 125-60-30 N–P_2O_5–K_2O; kg ha^{-1}.

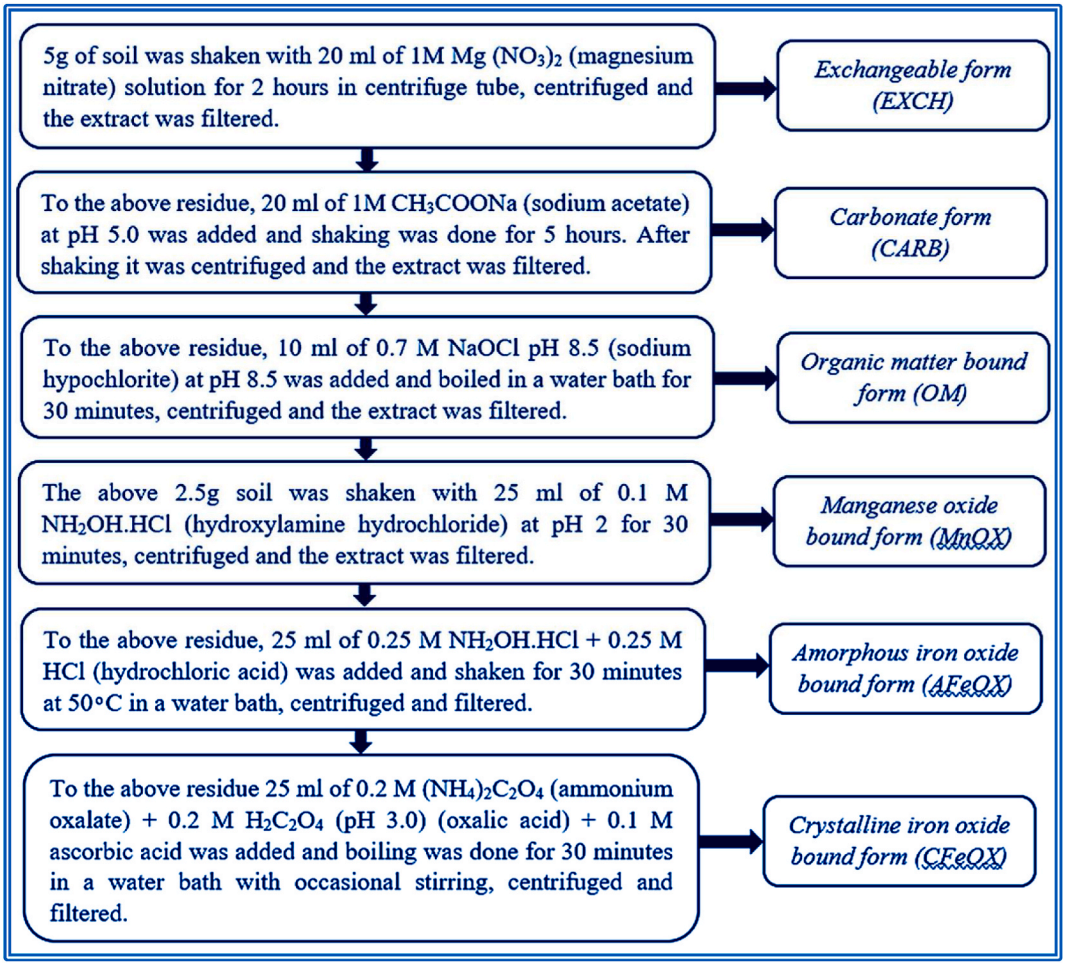


Fig. 1. Sequential fractionation procedure for determination of different fractions of Zn and Fe micronutrients.

which was true in the present investigation. Over the years, it was found to be lower (0.37 % in 2017-18) in the control treatment (T1) and higher in T6 (0.58 % in 2017-18) with the application of N + P + K + FYM (Table 3). This SOC content increased substantially up to the double extent throughout the experiment with slight fluctuations during 90s. It generated very valuable information which proved the notion wrong that soil fertility declined when the rice-wheat system was adopted. The inclusion of organic matter from FYM, WCS, and GM marginally boosted the soil's organic carbon content above that obtained with chemical fertilizer application. Long-term fertiliser use in conjunction with organic sources may have improved root development, resulting in more residue in the soil, which after decomposition may have raised the soil's organic carbon content [28]. The current finding confirmed the study of Bhattacharyya et al. [29]. According to Bhatt et al. [30], the application of N + P + K along with Zn and FYM resulted in the highest levels of OC in the surface and subsurface soil layers, whereas no fertiliser or FYM application significantly decreased the SOC content in control in both soil layers.

3.1.2. N, P, K and micronutrients

Soil N content showed an increasing trend (Tables 3 and 4) with time. It increased to a much higher level since the experiment was started. It was observed to be higher in T10 (195.70 kg ha^{-1}) treatment receiving green manuring followed by T11, T6, T7 and T9 during the current period. Application of 50 % NPK +50 % N substituted through GM or FYM to rice and 100 % NPK to wheat increased available N to the tune of 31.14 % over the control. Huang et al. [31] and Dhaliwal et al. [32] also observed higher N content in soil under organic manure than inorganic fertilizer application. In a long-term experiment, Upadhyay and Vishwakarma [33] observed that the available N content was higher by providing balanced nutrition to the soil under the rice-wheat cropping sequence, whereas a reduction in its content was noted under imbalanced nutrient management.

Unlikely, the value of available P showed fluctuations over the years. The treatments in which 50 % or 25 % nitrogen was replaced by farmyard manure, green manure or wheat cut straw showed an increased available P over the years in contrast to the treatments receiving chemical fertilizers alone. During 2017-18, the value of P ranged from 11.4 to 41.9 kg ha^{-1} and the minimum value of available P content was found in the control plot. Application of 50 % NPK and substitution of 50 % N through farmyard manure

Table 3
Soil fertility status after Rabi 2017–2018 from 0 to 15 cm depth under LTEs.

Treatments	pH	OC (%)				N (kg ha^{-1})				P (kg ha^{-1})				K (kg ha^{-1})			
	2017–18	1986–87	1996–97	2006–07	2017–18	1986–87	1996–97	2006–07	2017–18	1986–87	1996–97	2006–07	2017–18	1986–87	1996–97	2006–07	2017–18
T1	7.51^{b}	0.27^{e}	0.21^{f}	0.36^{e}	0.37^{f}	112^{g}	111.3^{e}	132.3^{h}	149.23^{e}	7.1^{i}	6.8^{j}	6.3^{j}	11.4^{j}	97^{g}	92.0^{d}	90.9^{h}	119.47^{d}
T2	7.81^{ab}	0.28^{e}	0.30^{e}	0.35^{e}	0.38^{f}	117^{g}	131.3^{d}	152.3^{g}	159.90^{d}	9.2^{h}	11.3^{i}	11.0^{i}	17.0^{i}	103^{ef}	108.0^{c}	93.6^{gh}	125.07^{c}
T3	7.60^{b}	0.32^{d}	0.30^{c}	0.36^{d}	0.40^{ef}	131^{e}	152.0^{c}	163.2^{f}	166.40^{d}	12.2^{e}	16.8^{i}	11.9^{i}	18.0^{i}	106^{e}	106.3^{c}	96.7^{g}	112.00^{e}
T4	7.67^{b}	0.33^{d}	0.31^{d}	0.36^{d}	0.37^{f}	125^{f}	149.3^{c}	163.2^{f}	172.80^{cd}	10.4^{g}	14.5^{h}	13.4^{h}	20.2^{h}	98^{g}	104.3^{c}	104.0^{f}	143.73^{a}
T5	7.63^{b}	0.35^{c}	0.32^{d}	0.37^{d}	0.41^{e}	148^{b}	170.7^{b}	170.2^{f}	176.37^{c}	13.7^{c}	17.0^{d}	24.3^{d}	28.5^{e}	107^{de}	106.3^{c}	110.8^{e}	121.33^{c}
T6	8.07^{a}	0.39^{a}	0.45^{a}	0.50^{b}	0.58^{a}	153^{b}	166.3^{b}	208.1^{b}	192.90^{a}	14.9^{a}	50.3^{a}	42.6^{a}	41.9^{a}	118^{bd}	117.7^{ab}	144.4^{a}	113.87^{e}
T7	7.63^{b}	0.37^{b}	0.40^{c}	0.49^{b}	0.56^{a}	142^{d}	155.3^{c}	195.1^{cd}	192.07^{a}	12.0^{c}	30.0^{b}	38.9^{b}	37.9^{b}	112^{cd}	111.5^{bc}	139.5^{b}	134.40^{b}
T8	8.02^{ab}	0.34^{c}	0.15^{g}	0.39^{c}	0.50^{c}	138^{cd}	165.7^{b}	198.1^{c}	189.40^{ab}	13.3^{cd}	18.8^{g}	18.3^{g}	25.1^{g}	114^{c}	112.0^{bc}	132.6^{c}	97.07^{f}
T9	7.62^{b}	0.34^{c}	0.45^{a}	0.38^{c}	0.47^{d}	139^{cd}	153.3^{c}	189.2^{d}	191.63^{ab}	12.0^{e}	16.6^{f}	20.5^{f}	27.1^{f}	104^{cf}	119.7^{a}	123.3^{d}	91.47^{g}
T10	7.67^{b}	0.39^{a}	0.40^{c}	0.39^{c}	0.56^{a}	151^{b}	167.0^{b}	217.1^{a}	195.70^{a}	15.0^{a}	21.9^{e}	22.5^{e}	30.5^{d}	120^{a}	115.0^{b}	132.7^{c}	78.80^{h}
T11	7.60b	0.38^{a}	0.42^{b}	0.38^{c}	0.54^{b}	146^{bd}	165.3^{b}	204.1^{bc}	193.37^{a}	12.8^{d}	17.1^{g}	19.1^{g}	28.1^{e}	110^{d}	112.0^{bc}	126.8^{d}	80.67^{h}
T12	7.73^{ab}	0.28^{e}	0.42^{b}	0.54^{a}	0.56^{a}	110^{h}	170.7^{b}	212.2^{ab}	184.50^{b}	7.1^{i}	36.5^{b}	38.5^{b}	33.6^{c}	101^{fg}	119.3^{a}	138.9^{b}	100.80^{f}
T13	7.70^{ab}	0.35^{c}	0.31^{d}	0.39^{c}	0.47^{d}	165^{a}	180.7^{a}	180.2^{e}	183.87^{bc}	11.2^{f}	11.9^{c}	27.5^{c}	31.3^{d}	115^{bc}	108.0^{c}	125.9^{d}	125.07^{c}
T14	7.97^{ab}	0.37^{b}	0.32^{d}	0.37^{d}	0.47^{d}	152^{b}	175.7^{ab}	174.1^{ef}	179.20^{bc}	14.3^{b}	14.3^{d}	25.2^{d}	28.6^{e}	117^{bc}	108.0^{c}	112.6^{e}	119.47^{d}
CD ($p \leq 0.05$)	0.35	0.01	0.01	0.01	0.02	5.63	6.99	8.59	7.65	0.52	1.16	1.13	1.28	3.72	4.44	3.60	5.46
The initial value (1983)	–	0.31				143.00				11.2				101.00			

Table 4
Soil fertility status (micro-nutrients) after Rabi 2017–2018 from 0 to 15 cm depth under LTEs.

Treatments	Zn (mg kg^{-1})			Cu (mg kg^{-1})			Fe (mg kg^{-1})			Mn (mg kg^{-1})		
	1986–87	2006–07	2017–18	1986–87	2006–07	2017–18	1986–87	2006–07	2017–18	1986–87	2006–07	2017–18
T1	1.70^{b}	1.44^{f}	1.3^{d}	0.78^{d}	0.70^{e}	2.1^{d}	10.2^{cd}	16.79^{i}	34.2^{f}	7.7^{cd}	7.03^{d}	17.2^{cd}
T2	1.75^{ab}	1.58^{e}	1.3^{d}	0.95^{b}	0.71^{e}	2.1^{d}	10.3^{cd}	24.74^{g}	33.5^{g}	7.6^{cd}	8.24^{c}	16.8^{d}
T3	1.76^{ab}	1.63^{de}	1.3^{d}	0.97^{ab}	0.75^{d}	2.7^{a}	10.5^{c}	21.35^{h}	40.5^{d}	7.7^{cd}	8.24^{c}	17.3^{cd}
T4	1.77^{ab}	1.64^{de}	1.3^{d}	0.96^{b}	0.76^{d}	2.6^{b}	9.9^{d}	28.77^{d}	37.0^{e}	8.6^{b}	8.31^{c}	16.8^{d}
T5	1.76^{ab}	1.66^{d}	1.7^{b}	0.96^{b}	0.81^{c}	2.6^{b}	11.3^{b}	27.63^{e}	36.4^{ef}	9.2^{a}	8.97^{b}	16.7^{d}
T6	1.79^{a}	1.94^{a}	1.7^{b}	1.01^{a}	0.90^{a}	2.3^{c}	11.3^{b}	32.48^{a}	35.6^{f}	9.5^{a}	9.24^{a}	16.7^{d}
T7	1.75^{ab}	1.82^{b}	1.8^{a}	0.90^{c}	0.79^{cd}	2.3^{c}	10.5^{c}	29.08^{cd}	45.3^{b}	9.2^{a}	8.77^{b}	17.8^{c}
T8	1.78^{ab}	1.76^{c}	1.8^{a}	0.93^{b}	0.88^{ab}	2.2^{c}	11.1^{b}	30.15^{bc}	44.1^{b}	9.5^{a}	8.46^{c}	18.0^{c}
T9	1.78^{ab}	1.69^{d}	1.4^{c}	0.97^{ab}	0.84^{bc}	2.1^{d}	10.6^{c}	29.98^{a}	42.8^{c}	8.6^{b}	8.27^{c}	16.7^{d}
T10	1.77^{ab}	1.79^{c}	1.4^{c}	0.95^{b}	0.78^{cd}	2.0^{d}	12.6^{a}	31.76^{ab}	41.3^{d}	9.5^{a}	8.80^{bc}	16.1^{d}
T11	1.77^{ab}	1.70^{d}	1.3^{d}	0.95^{b}	0.74^{de}	2.2^{c}	11.9^{b}	31.62^{ab}	47.5^{a}	8.7^{b}	8.79^{bc}	19.4^{ab}
T12	1.70^{b}	1.85^{b}	1.2^{e}	0.82^{d}	0.86^{b}	2.8^{a}	11.1^{b}	30.83^{bc}	41.3^{d}	7.9^{c}	9.40^{a}	19.0^{b}
T13	1.72^{ab}	1.49^{f}	1.3^{d}	0.92^{b}	0.76^{d}	2.5^{b}	10.6^{c}	26.10^{f}	42.1^{c}	7.4^{d}	9.14^{a}	20.3^{a}
T14	1.76^{ab}	1.66^{d}	1.3^{d}	0.95^{b}	0.69^{e}	2.6^{b}	12.4^{a}	22.06^{h}	37.4^{e}	8.9^{b}	9.31^{a}	19.6^{ab}
CD ($p \leq 0.05$)	0.08	0.06	0.05	0.04	0.03	0.11	0.44	1.09	1.47	0.44	0.38	0.90

resulted in a higher available P content of 41.9 kg ha^{-1} in comparison to 11.2 kg ha^{-1} at the initial stage of the experiment. Yaduvanshi et al. [34] also observed similar results.

In contrast to this, soil exchangeable K revealed the same trend as soil N and it increased with time (Table 3). The soil exchangeable K was higher in T4 (143.73 kg ha^{-1}). But soil available K status improved under the recommended doses of NPK fertilizers + organic manures to that of control as well as of their initial value (101.0 kg ha^{-1}). The results of LTE thus, clearly described the beneficial effect of the application of FYM and GM which improved the build-up of available nitrogen, phosphorus and potassium in soil when supplemented with inorganic fertilizers. The decline in yield trend at lower N levels over time reflects the soil's dwindling nutrient reserves [35]. Likewise, the micronutrients were also maximum in treatments receiving FYM, WCS and GM. Treatments T7 and T8 showed higher Zn (1.8 mg kg^{-1}) content, T12 Cu content (2.8 mg kg^{-1}) and T11 Fe (47.5 mg kg^{-1}) content in contrast to the inorganic source of nutrition (Table 4). These results are in accordance with the findings of Walia et al. [28] and Dhaliwal et al. [36].

3.1.3. Microbiological parameters

The microbial count is another important index of soil health and the data revealed that in rice higher bacterial population was observed in T7 and T9 (26.2×10^6) treatments and actinomycetes in T9 (34.4×10^4) and fungi in T5 (28.0×10^4) (Table 5). In wheat the maximum number of bacteria was found in T6 (28.8×10^6), actinomycetes in T12 (38.6×10^4) and fungi in T6 (29.6×10^3) Also, it was noted that the proportion of plots receiving higher potassium dressings had increased K availability in the treatments. In the case of rice, bacteria, actinomycetes and fungi in recommended fertilizer treatment were 23.9×10^6, 33.9×10^4 and 28.0×10^3, respectively while in wheat they were found to be 21.6×10^6, 34.8×10^4 and 25.8×10^3, respectively. Their population further reduces with lower levels of fertilizer treatments.

3.2. Effect of long-term INM on potassium dynamics in soils

3.2.1. Water soluble potassium

Water soluble K, which is a readily available source of potassium, may change owing to cropping or external supplies of potassium, such as inorganic fertilisers and other organic sources. Any change brought on by the uptake of K from crops is countered by the release

Table 5
Microbial counts of soils under integrated nutrient supplements in the Rice-Wheat system.

Treatments	Soil moisture (%)	Viable count (cfu g^{-1})		
		Bacteria ($\times 10^6$)	Actinomycetes ($\times 10^4$)	Fungi ($\times 10^3$)
Rice (2017-18)				
T1	9.0^{b}	18.0^{f}	28.1^{cd}	16.7^{d}
T2	9.2^{b}	19.1^{e}	29.5^{cd}	17.2^{d}
T3	9.2^{b}	24.8^{bc}	32.1^{b}	27.3^{ab}
T4	9.0^{b}	24.7^{bc}	34.3^{a}	26.5^{bc}
T5	9.2^{b}	23.9^{c}	33.9^{a}	28.0^{a}
T6	9.8^{a}	23.7^{c}	33.1^{ab}	27.3^{a}
T7	9.8^{a}	26.2^{a}	33.9^{a}	26.7^{b}
T8	9.6^{ab}	24.9^{b}	33.2^{ab}	26.2^{bc}
T9	9.6^{ab}	26.2^{a}	34.4^{a}	26.7^{b}
T10	9.8^{a}	19.9^{d}	30.3^{b}	25.8^{c}
T11	9.8^{a}	20.1^{d}	29.4^{cd}	16.9^{d}
T12	9.8^{a}	19.8^{d}	29.2^{cd}	16.8^{d}
T13	9.4^{ab}	20.1^{d}	29.0^{cd}	16.9^{d}
T14	9.2^{b}	19.4^{d}	28.9^{d}	16.8^{d}
CD ($p \leq 0.05$)	0.42	0.90	1.43	0.82
Wheat (2016-17)				
T1	9.6^{c}	21.8^{d}	34.4^{c}	25.0^{c}
T2	9.8^{bc}	21.0^{d}	34.0^{c}	25.6^{c}
T3	9.8^{bc}	20.6^{e}	34.2^{c}	25.4^{c}
T4	9.8^{bc}	20.4^{e}	34.6^{c}	25.6^{c}
T5	10.0^{b}	21.6^{d}	34.8^{c}	25.8^{c}
T6	10.2^{ab}	28.8^{a}	38.2^{a}	29.6^{a}
T7	10.0^{b}	27.8^{b}	37.8^{ab}	28.6^{ab}
T8	10.2^{ab}	26.2^{c}	36.8^{b}	29.4^{a}
T9	10.4^{a}	25.8^{c}	36.2^{bc}	28.0^{b}
T10	10.2^{ab}	26.4^{c}	36.6^{b}	28.8^{ab}
T11	10.0^{b}	25.6^{c}	36.0^{bc}	27.6^{b}
T12	10.0^{b}	28.4^{a}	38.6^{a}	28.8^{ab}
T13	9.6^{c}	21.8^{d}	35.0^{c}	26.0^{c}
T14	10.0^{b}	21.6^{d}	35.0^{c}	25.8^{c}
CD ($p \leq 0.05$)	0.37	0.97	1.46	1.11

Date of sampling and plating of rice: September 26, 2017 and September 27, 2017 and Date of sampling and planting of wheat: November 16, 2017 and December 13, 2017.
Note: Note: NS—non-significant; Letters means within columns followed by the same letter are not significantly different (LSD at $p = 0.05$).

of exchangeable K to soil solution because the water-soluble form of K is in dynamic equilibrium with exchangeable K. Water soluble K concentrations varied between 35.2 and 50.6 kg ha^{-1} under the various nutrient management techniques as depicted in Table 6, with T6 having the highest concentration and control having the lowest. Under the combined usage of inorganic and organic fertilizers, water-soluble K was likewise significantly higher than the control treatment. It was 50.6, 49.1 and 47.9 kg ha^{-1} in the treatments T6, T8 and T10 where 50 % of the recommended dose was applied through FYM, WCS and GM, respectively and was 8.1, 4.9 and 2.4 %, respectively, higher over T4 where 100 % of the NPK was applied through chemical fertilizers alone. This indicated that FYM proved more beneficial among various organic amendments and also over the use of 100 % inorganic fertilizers. These results showed that water-soluble K in the soil had a direct relationship with available K. A marked decrease in water-soluble K content in control (T1) treatment was also observed. This was due to more K uptake by plants. Vinutha et al. [37] also concluded similar results under the finger-millet cropping system. Water-soluble K content was observed highest in 50 % NPK through chemical fertilizers +50 % NPK through FYM revealing the beneficial impact of combined use of manures with chemical fertilizers for maintenance of K levels. It has also been observed that the water-soluble K content was proportionately higher in treatments receiving higher doses of potassium.

Following the wheat crop, potassium levels in soil varied from 31.9 to 49.2 kg ha^{-1} (Table 6), with T6 (50 % NPK + 50 % N through FYM to the preceding crop) having the highest levels. When 100 % of the recommended NPK was applied, T5 treatment which received only chemical fertilisers resulted in the highest water-soluble K content. Among INM treatments (T6 to T11), the application of 50 % nutrients through FYM (T6) recorded the highest water-soluble K content of 49.2 kg ha^{-1} followed by water-soluble K content of 48.5 kg ha^{-1} in T8 where 50 % of the substitution was done with WCS to the preceding rice crop. These values were higher than the water-soluble K content in T5, which received 100 % NPK through chemical fertilizers only. The plots getting 50 % of the recommended NPK through FYM had the highest levels of water-soluble K, possibly reflecting the benefits of FYM on the soil's characteristics and demonstrating FYM's superiority to the other organic sources [38]. Water-soluble K in treatments receiving 50 % and 25 % of the recommended NPK through FYM, WCS, and GM had higher levels than the treatment with 100 % chemical fertilisers (T5). The results thus, showed that FYM was the most effective organic amendment for raising water-soluble K.

Water-soluble potassium in surface layers has increased with the use of potassium fertilisers, weathering, organic manures, the release of water-soluble K from organic residues and the upward movement of K from deeper layers of soil with the rise of capillary water [39]. Using fertilizers along manures increased the amount of water-soluble K since manures improve the properties of soil [40]. The results of Lal et al. [41], Bhardwaj and Omanwar [42], Setia and Sharma [43], Singh et al. [44] and Talashilkar et al. [45] supported the conclusions of the current study.

3.2.2. *Exchangeable potassium*

Exchangeable K was highest in T6 in which 50 % of the recommended dose was applied through FYM measured at 111.1 and 112.8 kg ha^{-1} respectively, which is closely followed by T8, T7, T10 and T11 (Table 6). The reduced potassium fixation brought on by the frequent addition of manures combined with potassic fertilisers may be the reason for the increased exchangeable potassium under integrated nutrient supply over control. This clearly indicates that the integrated nutrient management treatments showed higher exchangeable potassium than chemical fertilizers alone treatments. Among organics, FYM proved to be the best amendment for

Table 6
Effect of long-term integrated nutrient management on fractions of potassium.

Treatments	Water soluble K	Exchangeable K	Available K	Non-exchangeable K	Fixed K	Total K
	After rice (kg ha^{-1})					
T1	35.2^{c}	37.6^{d}	72.8^{f}	1265.7^{e}	1338.5^{h}	11031.3^{f}
T2	39.8^{b}	41.4^{d}	81.2^{f}	1307.2^{e}	1388.4^{g}	11950.3^{e}
T4	41.1^{b}	61.9^{c}	103.1^{e}	1679.1^{d}	1782.2^{f}	12490.0^{d}
T5	46.8^{a}	70.8^{c}	117.6^{d}	1925.3^{c}	2042.9^{e}	13919.7^{c}
T6	50.6^{a}	111.1^{a}	161.6^{a}	2533.9^{a}	2695.5^{a}	15066.7^{a}
T7	48.6^{a}	103.9^{a}	152.6^{ab}	2387.9^{b}	2584.7^{c}	14916.7^{ab}
T8	49.1^{a}	87.8^{b}	136.9^{c}	2474.5^{a}	2611.4^{bc}	14751.3^{b}
T9	48.5^{a}	80.1^{bc}	128.6^{cd}	2384.2^{b}	2512.8^{d}	14697.3^{b}
T10	47.9^{a}	100.0^{a}	148.1^{b}	2486.4^{a}	2634.5^{b}	14992.0^{a}
T11	46.9^{a}	98.9^{ab}	145.8^{bc}	2394.2^{b}	2540.0^{cd}	14935.3^{ab}
CD ($p \leq 0.05$)	4.1	12.3	11.1	61.9	46.2	191.2
	After wheat (kg ha^{-1})					
T1	31.9^{d}	47.3^{e}	79.2^{f}	1285.1^{e}	1364.3^{e}	11019.3^{g}
T2	36.4^{c}	53.9^{e}	90.3^{e}	1291.0^{e}	1381.3^{e}	11075.2^{g}
T4	39.2^{c}	67.1^{d}	106.7^{d}	1553.0^{d}	1660.0^{d}	12079.9^{f}
T5	44.9^{b}	67.5^{d}	111.9^{d}	1880.7^{c}	1992.7^{c}	13032.2^{e}
T6	49.2^{a}	112.8^{a}	162.0^{a}	2369.4^{a}	2531.4^{a}	15072.2^{a}
T7	47.6^{ab}	103.7^{b}	151.3^{b}	2345.9^{ab}	2494.3^{ab}	14904.1^{ab}
T8	48.5^{ab}	93.5^{c}	142.0^{c}	2335.6^{ab}	2479.9^{b}	14508.4^{c}
T9	46.9^{ab}	87.9^{c}	134.9^{d}	2316.7^{b}	2456.0^{b}	14017.7^{d}
T10	48.2^{ab}	100.3^{bc}	148.5^{bc}	2340.9^{ab}	2485.8^{b}	14927.8^{ab}
T11	46.9^{ab}	97.8^{bc}	144.6^{c}	2325.2^{b}	2471.5^{b}	14775.2^{b}
CD ($p \leq 0.05$)	4.2	7.2	5.5	38.4	37.7	234.3

Note: Note: NS—non-significant; Letters means within columns followed by the same letter are not significantly different (LSD at $p \leq 0.05$).

increasing exchangeable K. This may be related to the possibility that adding FYM to the soil could boost its cation exchange capacity, causing a mass action effect by permitting the soil to hold more amount of exchangeable K and converting it from non-exchangeable to exchangeable form [46]. Similar results were also observed by Setia and Sharma [43], Singh et al. [44] and Talashilkar et al. [45].

3.2.3. Available K

The data on available K status revealed that in control plots, it was 72.8 kg ha^{-1} (Table 6). However, it significantly increased from 72.8 kg ha^{-1} in the control to 81.2, 103.1 and 117.6 kg ha^{-1} in the treatments T2, T4 and T5, respectively, where 50 %, 75 %, and 100 % of recommended doses of NPK were provided using chemical fertilisers. These results showed that the marked decline in available K content was due to more K uptake by plants. The decrease in available K content without K application was also reported by Biswas et al. [47] and Vinutha et al. [37]. Also, it was noted that the proportion of plots receiving higher potassium dressings had increased K availability.

The application of organic amendments significantly increased the amount of potassium that was readily available. T6 has the most available K (161.6 kg ha^{-1}), followed by T7, T8, and T10. In comparison to the treatment in which 100 % recommended dose of fertilizers was applied using inorganic fertilisers, the available K was 37.4, 16.4, and 25.9 % greater when 50 % of advocated NPK was applied through FYM, WCS and GM. Similar to the 100 % chemical fertiliser treatment, the available K was higher when 25 % of NPK was administered through FYM, WCS, and GM, coming in at 29.6, 9.35, and 23.9 %, respectively. In comparison to the treatment receiving 100 % NPK alone, Kher and Minhas [38] found that 100 % NPK + FYM treatments had higher accessible K values. This increase was attributed to potassium being applied at the optimal dose as well as through physical and biochemical benefits brought by the addition of FYM.

The amount of K that was readily available in the soil following wheat crop varied from 79.2 kg ha^{-1} under the control condition to 162.0 kg ha^{-1} under the treatment that received 50 % NPK from inorganic fertilisers along with 50 % recommended doses from FYM used on the previous rice crop as revealed from Table 8. Following the application of organic and inorganic fertilizers, the soil's accessible potassium content was significantly higher than the control. Under T6, the available potassium content increased by an average of 48.0 % compared to the control. The available potassium content was 162.0, 142.0 and 148.5 kg ha^{-1} where 50 % of recommended doses of fertilizers were supplied through FYM, WCS and GM, respectively to the previous rice crop. However, available K content in all these integrated treatments was significantly higher than it was in treatment T4 (111.9 kg ha^{-1}) where 100 % recommended doses of fertilizers were applied through chemical fertilizers only.

3.2.4. Non-exchangeable potassium

In integrated treatments (T6 to T11), treatment T6 recorded the highest non-exchangeable K after rice and wheat in which 50 % of the recommended dose of fertilizers was applied through FYM measured at 2369.4 and 2369.4 kg ha^{-1} respectively, followed by treatment T10 (2486.4 kg ha^{-1}) after rice in which 50 % of the recommended NPK was substituted through GM and T7 (2345.9) after wheat (Table 6). The non-exchangeable potassium was also relatively higher in treatments where 25 % of the substitution was done with FYM, WCS and GM, respectively in comparison to chemically treated treatments. Our findings demonstrated that organic manure addition along with chemical fertilisers considerably increased the non-exchangeable potassium compared to the control. Among organics, FYM recorded the highest increase in non-exchangeable potassium content after rice (31.6 %) and wheat (46.4 %) over T5 in which 100 % of the NPK fertilization was done with the chemical fertilizers alone. In addition to this, non-exchangeable potassium K was found more in K-treated plots than where K fertilizer was not added. This was due to the demand for plants which was met by added K fertilizers. This resulted in a reduced release of potassium from the non-exchangeable form. Depletion of non-exchangeable K was more in control plots because crops remove higher amounts of K from the soil. This clearly showed that non-exchangeable K played a vital role in meeting the K requirements of the crops. Ganeshamurthy [48] and Brar et al. [49] also observed that the addition of K through fertilizers limited its release from a non-exchangeable form.

The supply of potassium in the soil is continuously reduced when crops are produced one after another without potassium supplementation, increasing the demand for the nutrient. Due to the flow of potassium coming from a non-exchangeable form, the former drastically decreases in the dynamic equilibrium system [50]. Also, the crops' higher removal of potassium from non-exchangeable sources was a result of enhanced root proliferation in the top layer [43]. Additionally, the amount of non-exchangeable potassium surged as a result of integrated treatments. This may be due to organic manures being added repeatedly fixing the applied potassium balance as well as the residual potassium balance. Similar findings were reported by Santhy et al. [40], Bhardwaj and Omanwar [42] and Talashilkar et al. [45].

3.2.5. Fixed potassium

Fixed potassium in the soil after rice was highest (2695.5 kg ha^{-1}) in T6 treatment where 50 % of the recommended dose of fertilizers was supplied through farmyard manure (Table 6). The amount of fixed potassium was found to be 101.3, 95.0, and 96.8 per cent higher in treatments in which 50 % of the recommended dose of fertilizers was provided by FYM, WCS and GM, respectively over control. However, it was higher by the magnitude of 31.9, 27.8 and 28.9 per cent in treatments where 50 % of the recommended dose of fertilizers was supplied through FYM, WCS and GM over T5 in which 100 % of NPK fertilization was done through chemical fertilizers. Similarly, fixed K was higher by 26.5, 23.0 and 24.3 per cent in the treatments where 25 % of the recommended doses of fertilizers were supplied through FYM, WCS and GM, respectively over T5. A similar trend was followed after wheat i.e., the fixed potassium was highest when FYM was applied in conjunction with chemical fertilizers (Table 6).

The FYM treatment may have more fixed K because FYM enhances CEC, which holds more fixed K as a result of the mass action effect. The distribution of particle sizes, quantities and types of clay minerals, and K removal from minerals affect how much fixed K is

present in soil. It considerably correlates with the soil's silt and clay content [51]. The fixed levels of potassium in soils indicate that these soils are a source of potassium for plants and that illites, micas and other K-bearing minerals would quickly and irreversibly weather away, degrading the soil. For a steady supply of K to plants, fixed potassium levels in soils rich in minerals bearing potassium are essential. In the Indo-Gangetic plain, Mukhopadhyay and Datta [52] foresaw this scenario developing in many locations and recommended applying K dressing to make up for soils' K loss.

3.2.6. Total potassium

The major portion of the total potassium in soil is not available for plant uptake since it is a structural part of soil minerals. The total K changes were according to kaolinitic acidic alluvial soils < laterite soils < kaolinitic red < vertic intergrades < smectitic vertisols < illitic alluvial soils. The trends in the content of total potassium were comparable to those in fixed potassium. Total K in the present soil varied from 11031.3 to 15066.7 kg ha^{-1} soil under various treatments. The total potassium content significantly increased as fertiliser use increased (Table 6). In the treatment, wherein 100 % of the recommended dose of fertilizers was provided through inorganic fertilisers, the total K was 13919.7 kg ha^{-1}. Nevertheless, in the treatments when 50 % of the recommended NPK was delivered through FYM, WCS and GM, the potassium content was 15066.7, 14751.3, and 14992.0 kg ha^{-1}, respectively. With annual organic manure inclusion at 50 and 25 % compared to no organic assimilation, the concentration of total potassium increased considerably. Given that organic manures typically contain 1.4 % potassium, annual application over the past 32 years may have led to a build-up of total potassium in the soils. In comparison to treatment T5, where 100 % of the NPK fertilisation was carried out using chemical fertilisers only, FYM exhibited the highest increase in total potassium content (8.2 %). Similar findings were revealed by Dhanorkar et al. [53] in a study on the influence of the long-term application of FYM and NPK on different forms of soil K in vertisol. They noted distinct built-ups in total K were where K was applied and that a 40 % increase in total K was through FYM alone. A similar trend of higher total potassium in soil was applied with FYM in addition to chemical fertilizers (Table 6). After wheat, the total K content in treatment T6 was significantly higher by 36.7 % over control. The increase in total K content in treatments T6, T8, and T10 was 15.6, 11.3 and 14.5 % respectively over the T5 treatment in which 100 % NPK was applied through inorganic fertilizers only. The same trend as that of other forms of K was followed by total K.

The increase in total K may be a result of clay minerals fixing more soluble K. According to Yaduvanshi and Swarup [54], the soil's total K content showed a negative balance when 100 % NPK was applied, but there were gains over its initial status when 50 % NPK along with FYM was added. However, Singh et al. [55] discovered that the total potassium balance in the soil had decreased. The greater fall in total potassium in treatments that also received chemical fertiliser and organic manure was caused by the release of more potassium from the non-exchangeable potassium pool and subsequent uptake by crops. The upper layers, where the rooting density is highest, may have seen the greatest stress on K reserves to meet the K requirements of the crop.

3.3. Effect of long-term INM on zinc dynamics in soil

Among different Zn fractions in the surface soil (0–15 cm), the highest content of Zn in soil was found in crystalline iron oxide bound fraction (CFeOX) which varied from 1.64 to 3.08 mg kg^{-1} for different treatments and a significant variation existed among the

Table 7
Effect of integrated nutrient management on different Zn and Fe fractions (mg kg^{-1}) in surface soil (0–15 cm) under rice-wheat cropping system.

Treatments	ECXH	CARB	OM	MnOX	AFeOX	CFeOX
Zn fractions (mg kg^{-1})						
T1	0.60^{bc}	0.74^{b}	0.43^{bc}	1.49^{a}	1.02^{c}	2.42^{c}
T2	0.50^{c}	0.57^{c}	0.40^{bc}	0.96^{a}	1.02^{c}	2.92^{ab}
T4	0.53^{c}	0.59^{c}	0.35^{c}	1.49^{a}	1.63^{a}	1.64^{e}
T5	0.55^{c}	0.62^{c}	0.42^{bc}	1.43^{a}	1.16^{b}	2.83^{b}
T6	0.84^{a}	0.89^{a}	0.69^{a}	1.32^{a}	1.03^{bc}	2.23^{cd}
T7	0.79^{ab}	0.85^{ab}	0.63^{a}	1.44^{a}	0.94^{c}	3.08^{a}
T8	0.66^{bc}	0.75^{b}	0.46^{b}	1.13^{a}	0.94^{c}	2.88^{b}
T9	0.62^{bc}	0.71^{b}	0.45^{b}	1.09^{a}	0.95^{c}	2.76^{b}
T10	0.74^{ab}	0.81^{ab}	0.53^{b}	1.14^{a}	0.56^{d}	2.08^{d}
T11	0.69^{b}	0.79^{a}	0.51^{b}	1.12^{a}	0.57^{d}	2.77^{b}
CD ($p \leq 0.05$)	0.11	0.09	0.08	NS	0.15	0.16
Fe fractions (mg kg^{-1})						
T1	11.9^{bc}	20.6^{a}	0.36^{ab}	73.8^{c}	212.7^{d}	801.6^{bc}
T2	10.2^{c}	20.2^{a}	0.29^{b}	59.2^{d}	195.5^{d}	803.4^{bc}
T4	11.3^{c}	20.1^{a}	0.28^{b}	74.0^{c}	222.7^{cd}	717.1^{c}
T5	10.7^{c}	19.5^{a}	0.29^{b}	65.7^{cd}	310.5^{a}	876.5^{b}
T6	18.9^{a}	26.7^{a}	0.40^{a}	88.8^{b}	238.0^{c}	718.2^{c}
T7	18.3^{ab}	25.1^{a}	0.45^{a}	102.9^{a}	206.1^{d}	854.4^{b}
T8	15.0^{b}	23.5^{a}	0.35^{ab}	74.1^{c}	221.1^{cd}	984.8^{a}
T9	14.7^{bc}	23.0^{a}	0.37^{ab}	72.8^{c}	260.8^{bc}	767.0^{c}
T10	18.2^{ab}	25.9^{a}	0.43^{a}	85.3^{b}	280.8^{b}	809.2^{bc}
T11	17.9^{ab}	25.0^{a}	0.41^{a}	67.1^{cd}	242.7^{c}	609.3^{d}
CD ($p \leq 0.05$)	3.5	NS	0.10	9.8	22.8	76.2

e: NS—non-significant; Letters means within columns followed by the same letter are not significantly different (LSD at $p = 0.05$).

different treatments (Table 7). The treatments with the highest concentrations of exchangeable (EXCH), carbonate (CARB) and organic matter (OM) bound Zn were those that received 25 % and 50 % of the suggested NPK through FYM, WCS and GM, respectively. In the treatments receiving 50 % of the recommended dose of fertilizers through FYM, WCS and GM, EXCH-Zn was 0.84, 0.66, and 0.74 mg kg^{-1}, respectively and was 52.7, 20.0, and 34.5 %, respectively, greater than the treatment where 100 % of the recommended dose of fertilizers was supplied through inorganic fertilisers alone. As compared to the treatment receiving 100 % of the recommended dose of fertilizers, CARB-Zn was 43.5, 20.1, and 30.6 % more in treatments in which 50 % of the recommended dose was supplied through FYM, WCS and GM. Among various organic sources of nutrition, FYM treatments were found to contain the highest concentrations of EXCH, CARB and OM-Zn, followed by GM and WCS, respectively. In FYM treatments, the EXCH, CARB, and OM-Zn ranged from 0.79 to 0.84, 0.85 to 0.89 and 0.63–0.69 mg kg^{-1} soil, respectively. Dhaliwal et al. [56], reported significant effects of different manures on Zn and Fe transformation when rice was grown on coarse-textured soils. According to Shuman [57], the inclusion of organic amendments increased Zn in the organic, exchangeable and manganese-bound fractions. The addition of organic manures may have caused the release of more soluble forms of zinc in which the metal was bound by oxides [58]. Organic molecules present in FYM and GM may have released zinc that was bound by crystalline oxides and carbonates [59]. The increase in concentrations of water-soluble plus exchangeable, OM-bound, manganese oxide bound, and amorphous iron oxide bound (AFeOX) bound fractions of Zn with the application of organic manures over their initial values was also observed by Sekhon et al. [60]. According to Mandal et al. [61], the application of 0.5 % FYM resulted in a significant rise in the AFeOX bound fraction of native soil Zn content and a concurrent decrease in CFeOX bound and Al oxides bound Zn fractions. An increase in Zn content in MnOX and AFeOX bound fractions at the expense of other fractions especially the CFeOX bound fractions with the addition of organic matter was also reported by Shuman [62]. Under the pearl-millet wheat system, Narwal et al. [63] also investigated the impact of FYM on Zn fractions. They claimed that when FYM is added, all Zn fractions increase, demonstrating the significance of organic matter addition. The addition of organic matter causes Zn to migrate from less soluble forms to a fraction that is more readily available to plants as reported by Ref. [64].

3.4. Effect of long-term INM on iron dynamics in soil

Among different fractions of Fe, the dominating fraction was CFeOX in the surface soil and its content ranged between 717.1 and 984.8 mg kg^{-1} followed by AFeOX-Fe which ranged between 195.5 and 280.8 mg kg^{-1} for all the treatments (Table 7). The treatments in which 50 and 25 % of the recommended dose of fertilizers were administered by FYM, WCS and GM, respectively, had the highest levels of EXCH, CARB, and OM bound Fe and these treatments showed 76.8, 40.1 and 70.2 %, respectively higher EXCH Fe than the treatment where 100 % of the advised dose of fertilizers were applied using chemical fertilisers alone. Treatments receiving FYM were found to have the highest concentrations of EXCH, CARB, and OM-Fe, followed by GM and WCS, respectively.

Farmyard manure applied field had much more soluble Fe than other nutrient management approaches, according to Agbenin and Henningsen [65]. This could be attributed to the mobilisation of Fe through the production of soluble Fe-organic complexes with the addition of organic matter. Similar results were reported by Sekhon et al. [60] in the rice-wheat sequence and Behera et al. [66] in the maize-wheat sequence. Yadav and Yadav [67] observed the effect of different levels of FYM and iron addition on iron transformation under graded levels of alkalinity. The content of different Fe fractions in soil decreased significantly with an increase in alkalinity but increased with the addition of FYM. The highest exchangeable Fe, adsorbed Fe, occluded Fe and organically bound Fe were recorded under the lowest level of alkalinity and the highest level of FYM. Dhaliwal et al. [68] also reported significant changes in Fe content in its different fractions when different organic sources were added in combination with different levels of chemical fertilisers.

3.5. Correlation between soil parameters

The correlation matrices pertaining to important characters viz., average yield (rice, wheat) and soil fertility status after rabi 2017-18 (pH, OC, N, P, K and micronutrients) are given in Tables 8 and 9. The data thus clearly revealed a strong and positive correlation between crop yields and soil properties (N, P and OC) but a negative correlation with K indicating that with an increase in crop yields both in rice and wheat, the soil is becoming deficient in K and requires the application of potassium fertilizer. The results are in agreement with the study of Bhunia et al. [69] who stated that OC was significantly and positively correlated with N. Similarly, crop yields (rice and wheat) showed a significantly strong correlation with N (Table 8). So far, the availability of the micronutrients concerned in a long-term experiment of rice-wheat system micro-nutrient contents is decreased. Cu was negatively correlated with Zn, Fe and Zn showed a negative correlation with Cu and Mn (Table 9) indicating the soils are becoming deficient in micro-nutrients over time and require their application to the soil to keep the threshold level intact.

Table 8
Pearson correlation coefficients of soil properties in 2017 with 34 years average yield in the rice-wheat long-term experiment.

	Rice average yield	Wheat average yield	OC	N	P	K
OC	0.677* (0.007)	0.574* (0.031)	1.00	–	–	–
N	0.872* (<0.000)	0.785* (0.009)	0.860* (<0.000)	1.00	–	–
P	0.810* (0.000)	0.756* (0.001)	0.861* (<0.000)	0.832* (0.000)	1.00	–
K	−0.266 (0.357)	−0.161 (0.528)	−0.512 (0.061)	−0.499 (0.068)	−0.146 (0.617)	1.00

Note: Figures in parenthesis represent p-value and * represents significance at a 5 per cent level of significance.

Table 9
Pearson correlation coefficients of soil properties (micro-nutrients) in 2017 with 34 years average yield in the rice-wheat long-term experiment.

	Rice average yield	Wheat average yield	pH	Zn	Cu	Fe	Mn
pH	0.228 (0.431)	0.398 (0.158)	1.00	0.361 (0.204)	0.029 (0.920)	−0.171 (0.557)	0.116 (0.692)
Zn	0.259 (0.369)	0.356 (0.210)	0.361 (0.204)	1.00	−0.209 (0.472)	0.160 (0.583)	−0.292 (0.309)
Cu	0.175 (0.548)	0.307 (0.285)	0.029 (0.920)	−0.209 (0.472)	1.00	−0.073 (0.803)	0.385 (0.173)
Fe	0.564 (0.035)*	0.458 (0.099)	−0.171 (0.557)	0.160 (0.583)	−0.073 (0.803)	1.00	0.421 (0.133)
Mn	0.343 (0.228)	0.215 (0.458)	0.116 (0.692)	−0.292 (0.309)	0.385 (0.173)	0.421 (0.133)	1.00

Note: Figures in parenthesis represent p-value and * represents significance at a 5 per cent level of significance.

4. Conclusions

Soil organic carbon and its available nutrient status (nitrogen, potassium except for phosphorus) and zinc after Rabi 2017-18 in 0–15 cm depth are positively correlated with crop yields. There has been a depletion of potassium in soil over the years and it required the application of K fertilizer. The treatments receiving organic manures showed a discernible increase of soil microorganisms in comparison to those receiving chemical fertilizers, thereby, indicating the superiority of these systems over chemical treated treatments. Among different fractions of Zn and Fe, their highest content was found in crystalline iron oxide bound fraction and there existed a significant difference among different treatments. The highest amount of exchangeable, carbonate and organic matter-bound zinc and iron were found in treatments in which 50 % and 25 % of the recommended dose of fertilizers were applied through farmyard manure, wheat cut straw and green manure respectively, thus solving the superiority and synergistic effect of integrated nutrient management. The results thus, amply elucidate the importance of integrated nutrient management towards restoration of soil health along with the increase in productivity and further improvement of the quality characteristics. This approach will keep the food security of staple food intact. The supplementation of chemical fertilizers with green manuring after harvesting of wheat/or the use of excessive wheat straw/or a small quantity of farmyard manure seems a priority in the Indo-Gangetic Plains comprising 10.8 m ha area to maintain the buffer stock of the country and meet the eventuality on account of weather aberration/climate change in the coming time.

Funding

The study was funded by the Indian Council of Agriculture Research through All India Coordinated Research Project on Integrated Farming System operated under ICAR-Indian Institute of Farming System Research and Punjab Agricultural University, Ludhiana, India. The study was also financially supported by the Slovak Research and Development Agency under contract no. APVV-15-0562, APVV-20-0071 and Science grant agency under contract no. VEGA- 1/0300/22. The researchers would also like to acknowledge the Deanship of Scientific Research, Taif University, Taif, Saudi Arabia for funding this work.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Sohan Singh Walia: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Salwinder Singh Dhaliwal:** Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Roopinder Singh Gill:** Writing – original draft, Visualization, Validation, Resources, Methodology, Conceptualization. **Tamanpreet Kaur:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Karmjeet Kaur:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Mehakpreet Kaur Randhawa:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Oliver Obročník:** Writing – review & editing, Software, Resources, Project administration, Funding acquisition, Formal analysis, Data curation. **Viliam Bárek:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Data curation. **Marian Brestic:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Data curation. **Ahmed Gaber:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Data curation. **Akbar Hossain:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the Indian Council of Agricultural Research for funding the All India Coordinated Research Project on Integrated Farming System operated under ICAR-Indian Institute of Farming System Research and Punjab Agricultural University, Ludhiana, India for facilities and support. The authors are also thankful to the Slovak Research and Development Agency, Slovakia under contract no. APVV-15-0562, APVV-20-0071 and Science Grant Agency under contract no. VEGA- 1/0300/22 for supporting the study. The researchers would also like to acknowledge the Deanship of Scientific Research, Taif University, Taif, Saudi Arabia for funding this work.

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